

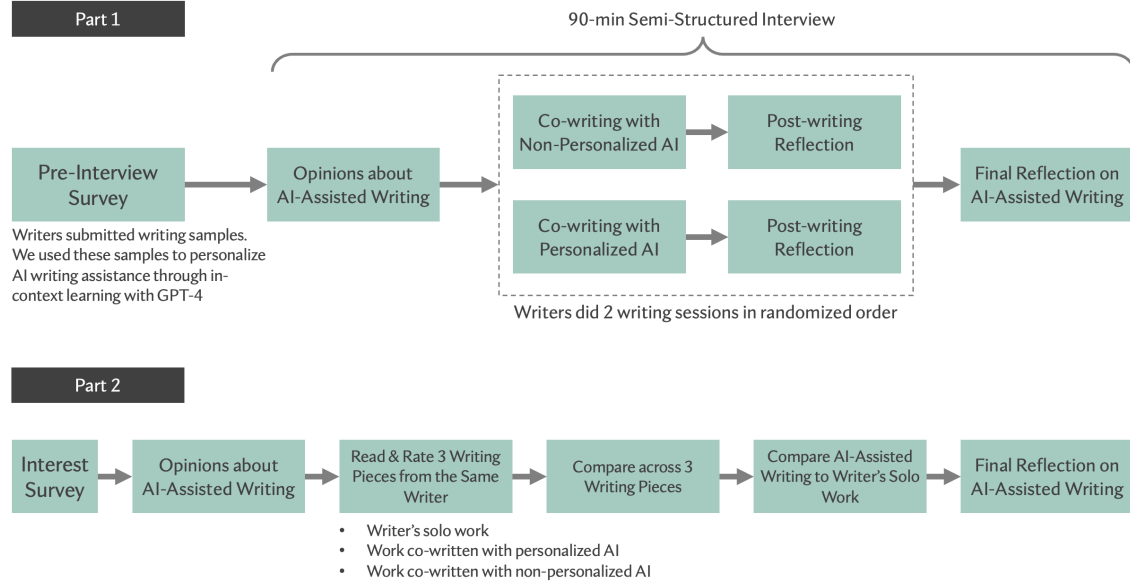


Fig. 1. Study procedure for Part 1 (interview study with writers) and Part 2 (survey study with readers).

3.2.1 Recruitment and participants. We recruited 21 professional writers through Upwork, a platform frequently used to recruit professionals with specific expertise. We partnered with a recruitment specialist from the Upwork team to ensure candidates provided authentic information on their profiles and verified their past freelance work on the platform. We also used the pre-interview survey for screening purposes, in which participants elaborated on their professional experiences as writers, shared their writing samples, and provided relevant links to their professional websites and profiles (see more details of the pre-interview survey in Section 3.2.2). One of the participants did not show up and one decided to drop out from the study, resulting in $N = 19$ participants who fully completed the study; data from all 19 participants was used for analyses. In Table 2, we report the background and experiences of these writer participants.

A note on sample size: Like the majority of qualitative research, we did not have a definite approach to determining the exact sample size. However, we followed recommendations from prior methodological reviews [6, 46] such that (1) we ensured the number of interviews conducted for the same task fell between 15 – 30; (2) we referred to the sample sizes of qualitative studies in relevant publication venues (e.g., [17, 18, 66]); and (3) throughout the study period, we observed that emergent themes arose and saturated even accounting for new interviews. Despite the relatively small sample size, we ensured that participants had diverse backgrounds to improve the generalizability of our results, such that our findings were not confounded with participants' seniority, experience, and writing genre.

3.2.2 Pre-interview survey. The pre-interview survey begins with obtaining informed consent from participants (by signing a consent form). After consenting, participants responded to a series of questions through free text responses. These questions ask participants to describe (1) the unique characteristics of their writing, (2) their prior experience working with AI and non-AI writing support tools, if any, (3) their definition of authenticity in writing. At the end of the survey, participants were asked to submit a short writing sample of their own (200 words) that best represents

their writing. Participants were informed their writing samples would be used as to design the study task for them (if selected to participate), though we did not specify how so. These writing samples were used for in-context learning to prepare a personalized AI writing assistance tool for the interview session (see more details in the next section), and a subset were used in the online survey with readers (see Part 2 Method in Section 3.3).

3.2.3 AI writing assistance. We adopted open-source code from [40] to run the CoAuthor interface (see Figure 2) with GPT-4 and used it as the AI writing assistance tool for the interview sessions. CoAuthor is a system built by Lee et al. [40] that allows writers to request next-sentence suggestions from an LLM, and the open-source code allows for customization, such as choosing which LLM to use and its parameters. We chose CoAuthor as the writing assistance

Participant	Primary Writing Genre	Years of Experience	Experience with genAI
P1	Science fiction, fantasy	5 - 10	P1 regularly used genAI tools but only for text editing purposes.
P2	Comedy, science fiction	10 - 15	P2 frequently used genAI tools for a wide range of purposes and was familiar with genAI models, such as GPT-3 and GPT-4.
P3	Lifestyle	<5	P3 regularly used genAI tools for text editing purposes (specifically for personal blog posts). They occasionally used genAI to brainstorm ideas for titles of their written work.
P4	Science writing	<5	P4 regularly used genAI tools but only for text editing purposes.
P5	Fantasy	<5	P5 had no experience with genAI at the time of the study.
P6	Lifestyle	5 - 10	P6 regularly used genAI tools but only for text editing purposes.
P7	History	15 - 20	P7 regularly used genAI tools to rewrite and make their drafts clearer.
P8	Science fiction	10 - 15	P8 had no experience with genAI at the time of the study.
P9	Lifestyle	<5	P9 regularly used genAI tools but only for text editing purposes.
P10	Biography	>20	P10 had no experience with genAI at the time of the study.
P11	Science writing	5 - 10	P11 had experimented with and explored the capabilities of various genAI tools. They did not find these tools useful for their writing jobs and thus did not use them regularly.
P12	Poetry	>20	P12 regularly used genAI tools to conduct research for their writing and used them for text editing purposes.
P13	Horror	10 - 15	P13 occasionally used genAI tools to write informational content.
P14	Romance	5 - 10	P14 used genAI tools for general purposes but not for writing.
P15	Spiritual	5 - 10	P15 had no experience with genAI at the time of the study.
P16	Romance	5 - 10	P16 regularly used genAI tools but only for text editing purposes.
P17	Science fiction	<5	P17 regularly used genAI tools but for text editing purposes; such practice was requested by many of their clients. For a specific client, the participant was provided with informational content generated by genAI and was asked to provide "human voice" to the writing.
P18	Fantasy	5 - 10	P18 frequently used genAI tools to generate written content for a client.
P19	Screenplay	>20	P19 regularly used genAI tools to generate visual content accompanying their writing work. They also frequently used genAI tools to help with formatting their writing.

Table 2. Professional profiles of writer participants in Part 1

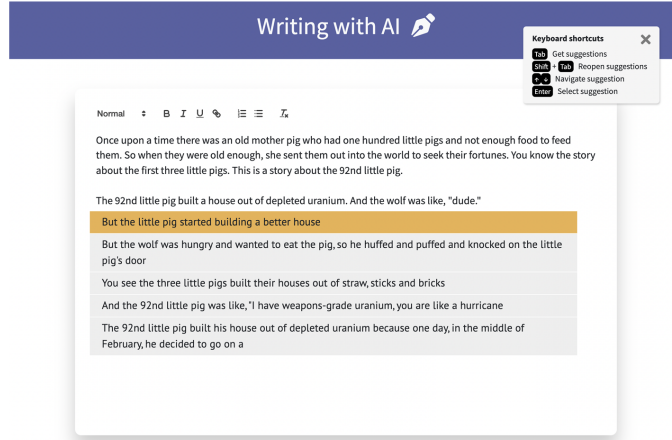


Fig. 2. Interface of the CoAuthor system (image adopted from [40]), which contains a text editor where users can request (Tab key) and adopt AI (Enter key) suggestions through keyboard shortcuts.

tool for this study for two reasons: (1) As we targeted writers with expertise in a diverse set of writing genres, we chose a tool that was *not* designed to support a particular type of writing. (2) Since participants had a limited time to become familiar with the AI writing assistance, we avoided more complex tools with multiple forms of support (e.g., Wordcraft [66]).

Before each session, we used the writer’s submitted writing sample as training data for in-context learning to create a personalized version of the CoAuthor tool for their session. We used the following parameters for GPT-4 to generate suggestions: max tokens = 5; temperature = 0.9; top p = 1; presence penalty = 0.5; frequency penalty = 0.5. During the co-writing sessions, writers were able to request AI writing suggestions at any time by the Tab key. The tool would generate around 5 sentence-level suggestions based on the written text. Writers could accept, reject, and revise the suggestions as they liked. At the beginning of their first writing session, the researcher who led the study session walked the writers through the CoAuthor interface and guided them to practice using the tool to write a few sentences. This was to ensure participants fully understood and became familiar with the tool before they started their first formal writing session.

3.2.4 Co-writing session with AI. Participants were asked to write one short passage (~200 words) for each co-writing session; participants chose to write on a topic similar (but not identical) to that of their submitted writing sample. Each co-writing session lasted between 20 - 25 minutes. The time was determined by piloting with two professional writers. All participants were able to finish their assigned 200-word passage within the time frame. As participants wrote, the CoAuthor interface recorded their writing logs and their final writing output. Writers’ behavioral data recorded through the writing logs include (1) the frequency and timing of their AI suggestion requests, (2) the frequency and timing of their acceptances and/or rejections of AI suggestions, (3) the AI text suggestions provided at each of their requests, (4) the AI suggestions (in text) accepted, if any, and (5) the text inserted by writers. Additional details about the writing logs can be found in Appendix D. After each writing session, participants participated in a short interview (~15 minutes) to reflect on their writing experiences.